NEXUS Software — Functionality and Readiness Gap Report

Audit date: 28 October 2026
Scope: current repository, schema, routers, services, tests and existing verification logs.
Change policy: no application-data or feature fixes were applied as part of this report. The only code-level change in this documentation batch is removal of an incompatible dependency override that broke Express startup.

Executive assessment

NEXUS has a credible database-backed academic core: People, attendance, allocations, timetable generation/editing, assessment rules, permissions, authentication, school registry and audit services. Payroll and Transport have substantial service-level implementations. Several Finance & Resources screens are partial or presentational. Multi-school isolation has strong foundations and tests but is not yet SaaS-grade across every legacy path. External services are mostly links, queues or provider-ready infrastructure rather than live integrations.

Status legend

- **Working/verified:** source and targeted tests/logs support the behavior.
- **Partially working:** some real persistence or workflow exists, but important stages are missing.
- **Unverified:** code exists, but a complete live/browser acceptance path was not evidenced.
- **Missing/placeholder:** UI language exists without a complete backing workflow.
- **External configuration:** requires provider, secrets, deployment or owner policy.

Priority findings

Priority	Module/submodule	Finding	Status	Evidence/verification
Critical	Multi-school / Curriculum & settings	Some legacy reads/writes are not explicitly school-scoped; curriculum uses an unscoped settings limit and offerings query	Risk/partial	`server/curriculum.ts`; `docs/multi-school-security.md`
Critical	Multi-school / Global keys	Some uniqueness is global rather than `(schoolId,key)`	Risk	`drizzle/schema.ts` subjects, admissions/payment references
Critical	Account recovery	Forgot-password creates token but no mail/SMS delivery	External configuration	`server/iam.ts` source note
High	Finance / Fees	Fee-item CRUD, invoices, discounts, arrears approval and official receipts not verified	Partial	Finance workspace/service audit
High	Finance / Procurement	Purchase requests, quotations, POs, goods receipts and supplier invoices are UI descriptions/placeholders	Missing/placeholder	`ResourcesFinanceWorkspace.tsx` dedicated domain service found
High	Finance / Assets	No full asset register/lifecycle/depreciation/custody workflow	Partial/placeholder	Assets reuses stock-style rows
High	Finance / Maintenance	Maintenance form writes synthetic expenditure-like record, not work order	Partial/weak	Finance operations audit
High	Assessment / Report Cards	Demo fallback and hard-coded institution/contact/term content in report template	Risk	`IntegratedReportCardView.tsx`
High	Deployment	Windows/XAMPP installer and rollback/restore not executed end-to-end	Unverified	`installer/AUDIT-CHECKLIST.md`

Priority	Module/submodule	Finding	Status	Evidence/verification
Medium	Attendance	UI explicitly exposes Grade 7–9 selectors	Partial scope	`AttendanceWorkspace.tsx`
Medium	Academics	Assessment, report-card, archive and curriculum full browser acceptance logs are incomplete	Unverified	Unit tests exist; no equivalent live logs
Medium	Archives/Alumni	Exit requires raw learner ID; no verified archive/alumni export	Partial	`ArchivesCounsellingWorkspace.t`
Medium	Finance / Reports	Many named finance reports are buttons/toasts, not dedicated reports	Partial/placeholder	Finance component/source audit
Medium	Messaging	Messages queue records but no external delivery	Queue-only	`server/messages.ts`
Medium	AI / Amal	Amal is new-tab link; no embedding/API/SSO/file exchange; document format selector not proof of files	External handoff/partial	`Home.tsx`, AI workspace audit
Medium	AI / M-Pesa	Ledger is callback-ready, live gateway not verified	External configuration	AI workspace source
Low	Dependencies	Automatic audit remediation updated many packages; Express 4 broke when path-to-regexp was forced to 9	Fixed in this batch; residual audit remains	`pnpm why path-to-regexp`; tests passed after removal
Low	Audit trail	Application audit exists, but immutable retention/export/alerting not evidenced	Partial	`smis_audit_logs`, administration au
Low	Backup	Scripts/runbooks exist, but scheduling, object storage, alerts and restore drill are not configured/proven	Operational code only	`ops/`, `docs/backup-and-restore.md`
Low	XLSX import/export	Remaining `xlsx` advisory was not replaced in this scope	Residual risk	Dependency audit

What was actually verified

- `pnpm check` passed after the safe dependency repair.
- `pnpm test -- --run` passed: 18 test files and 80 tests in the current source state.
- Source-backed authentication, permissions, tenant acceptance tests, Resource Hub tests, curriculum tests, academic tests, attendance tests, People import tests, payroll and transport tests exist and pass in the recorded run.
- The Express startup regression was reproduced by the deployment log (`pathRegexp is not a function`) and corrected by removing the incompatible `path-to-regexp` override; dependency resolution now returns Express 4 → `path-to-regexp 0.1.12` .
- No production database records were changed by this documentation batch.

Features not to promise

Do not promise live SMS/email/WhatsApp, M-Pesa collection, bank transfer, statutory filing, KEMIS/KNEC/TPAD API integration, embedded Amal, offline operation, automatic backups, immutable audit, MFA, full production multi-school SaaS isolation or a tested Windows installer unless separately verified.

Approval sequence

1. Owner reviews this report and approves the remediation priorities.
2. Security/tenant fixes are implemented and tested before adding schools.
3. Finance domain gaps are resolved before marketing fees/procurement/assets as complete.
4. Report-card template and external-provider disclosures are corrected.
5. Windows/XAMPP and disposable restore acceptance is executed.
6. Updated manuals are re-audited against the new labels and evidence.